



PULMOVALL 300 mg/ml

Solution for injection

Florfenicol



QUALITATIVE AND QUANTITATIVE COMPOSITION

Each ml contains:

Florfenicol	300 mg
Propylene glycol	150 mg
N-methyl-2-pyrrolidone	250 mg
Macroglol 300	q.s.t 1 ml

PRODUCT DESCRIPTION

The veterinary medicine is presented as a non-aqueous solution for injection. The physicochemical characteristics are appropriate for the dosage form; the drug is presented as a sterile, clear, yellowish solution free of particles in suspension.

PHARMACODYNAMICS

Florfenicol is a synthetic broad spectrum antibiotic effective against most Gram-positive and Gram-negative bacteria isolated from domestic animals. Florfenicol acts by inhibiting protein synthesis at the ribosomal level and is bacteriostatic. Laboratory tests have shown that florfenicol is active against most commonly isolated bacterial pathogens involved in bovine (including *Pasteurella multocida*, *Mannheimia haemolytica* and *Histophilus somni*) and in swine respiratory disease (including *Actinobacillus pleuropneumoniae* and *Pasteurella multocida*). Although considered to be a bacteriostatic agent, bactericidal activity of florfenicol has been demonstrated in vitro against *Mannheimia haemolytica*, *Pasteurella multocida*, *Actinobacillus pleuropneumoniae* and *Histophilus somni*.

Mechanisms of resistance to florfenicol include specific and non-specific drug transporters and RNA methyltransferases. In general, the specific efflux proteins provide levels of resistance greater than that of the multidrug efflux proteins. A number of genes (including *floR* gene) mediate combined resistance to florfenicol. Resistance to florfenicol and other antimicrobials has been firstly detected on a plasmid in *Photobacterium damsela* subsp. *piscicida*, then as part of a chromosomal multi-resistance gene cluster in *Salmonella enterica* serovar *Typhimurium* and serovar *Agona*, but also on multi-resistance plasmids of *E. coli*. Co-resistance with the third generation cephalosporins has been observed in respiratory and digestive *E. coli*.

PHARMACOKINETICS

Cattle: Intramuscular administration at the recommended dose of 20 mg/kg maintains efficacious blood levels in cattle for 48 hours. Maximum mean serum concentration (C_{max}) of 3.37 µg/ml occurs at 3.3 hours (t_{max}) after dosing. The mean serum max concentration 24 hours after dosing was 0.77 µg/ml. Subcutaneous administration at the recommended dose of 40 mg/kg maintains efficacious blood levels in cattle (ie. above the MIC₉₀ of the main respiratory pathogens) for 63 hours. Maximum serum 90 concentration (C_{max}) of approximately 5 µg/ml occurs approximately 5.3 hours (t_{max}) after dosing. The mean serum concentration 24 hours after dosing is approximately 2 µg/ml. The harmonic mean elimination half-life was 18.3 hours.

Pigs: After intravenous administration florfenicol had a mean plasma clearance rate of 5.2 ml/min/kg and a mean volume of distribution at equilibrium of 948 ml/kg. The mean terminal half-life is 2.2 hours.

After initial intramuscular administration of florfenicol, maximum serum concentrations of between 3.8 and 13.6 µg/ml are reached after 1.4 hours and the concentrations deplete with a terminal mean half-life of 3.6 hours. After a second intramuscular administration, maximum serum concentrations of between 3.7 and 3.8 µg/ml are reached after 1.8 hours. Serum concentrations drop below 1 µg/ml, the MIC₉₀ for the target porcine pathogens, 12 to 24 hours following IM administration. Florfenicol concentrations achieved in lung tissue reflect plasma concentrations, with a lung: plasma concentration ratio of approximately 1. After administration to pigs by the intramuscular route, florfenicol is rapidly excreted, primarily in urine. The florfenicol is extensively metabolised.

INTERACTIONS WITH OTHER MEDICAMENTS

None known.

INDICATIONS FOR USE

Diseases caused by florfenicol susceptible bacteria.

Cattle: Metaphylactic and therapeutic treatment of respiratory tract infections in cattle due to *Mannheimia haemolytica*, *Pasteurella multocida* and *Histophilus somni*. The presence of the disease in the herd should be established before metaphylactic treatment.

Pigs: Treatment of acute outbreaks of swine respiratory disease caused by strains of *Actinobacillus pleuropneumoniae* and *Pasteurella multocida*.

CONTRAINDICATIONS

- Do not use in adult bulls and boars intended for breeding purposes.
- Do not use in the case of known hypersensitivity to the active substance or to any of the excipients.

ADVERSE REACTIONS

Cattle: A decrease in food consumption and transient softening of the faeces may occur during the treatment period. The treated animals recover quickly and completely upon termination of treatment. Intramuscular and subcutaneous administration may cause inflammatory lesions at the injection site which may persist for 14 days. In very rare cases, anaphylactic shock has been reported in cattle.

Pigs: Commonly observed adverse effects are transient diarrhoea and/or peri-anal and rectal erythema/oedema which may affect 50% of the animals. These effects can be observed for one week. Under field conditions approximately 30% of treated pigs presented with pyrexia (40°C) associated with either moderate depression or moderate dyspnoea a week or more after administration of the second dose. Transient swelling lasting up to 5 days may be observed at the site of injection. Inflammatory lesions at the injection site may be seen up to 28 days.

The frequency of adverse reactions is defined using the following convention:

- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

USE DURING PREGNANCY, LACTATION OR LAY

Studies in laboratory animals with florfenicol have not revealed any evidence of teratogenic or fetotoxic effects. Laboratory



BARCODE





studies with the excipient N-Methylpyrrolidone in rabbits and rats have shown evidence of teratogenicity, fetotoxicity and maternal toxicity.

Cattle: the effect of florfenicol on bovine reproductive performance and pregnancy has not been assessed. Use only according to the benefit/risk assessment by the responsible veterinarian.

Pigs: the safety of the product in sows during pregnancy and lactation has not been demonstrated. Do not use the product during pregnancy and lactation.

TARGET SPECIES

Cattle and pigs.

DOSAGE FOR EACH SPECIES, ROUTES AND METHOD OF ADMINISTRATION

Administration route: intramuscular and subcutaneous in cattle, and intramuscular in pig.

For treatment:

Cattle: Intramuscular use: 20 mg/kg bodyweight (1 ml/15 kg) to be administered twice 48 hours apart. Subcutaneous use: 40 mg/kg bodyweight (2 ml/15 kg) to be administered once only.

For both routes: use a 16-gauge needle. The dose volume given at any one injection site should not exceed 10 ml. The injection should only be given in the neck.

Pig: Intramuscular use: 15 mg/kg bodyweight (1 ml/20 kg) by intramuscular injection into the neck muscle twice at 48-hour intervals using a 16-gauge needle. The volume administered per injection site should not exceed 3 ml. It is recommended to treat animals in the early stages of disease and to evaluate the response to treatment within 48 hours after the second injection. If clinical signs of respiratory disease persist 48 hours after the last injection, treatment should be changed using another formulation or another antibiotic and continued until clinical signs have resolved.

For metaphylaxis:

Cattle: Subcutaneous use: 40 mg/kg bodyweight (2 ml/15 kg) to be administered once only using a 16-gauge needle. The dose volume given at any one injection site should not exceed 10 ml. The injection should only be given in the neck.

For all target species: Wipe the stopper before removing each dose. Use a dry, sterile syringe and needle. To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.

OVERDOSE

Cattle: No symptoms.

Pig: After administration of 3 times the recommended dose or more a reduction in feeding, hydration and weight gain has been observed. After administration of 5 times the recommended dose or more vomiting has also been noted.

WITHDRAWAL PERIODS

Meat and offal: **Cattle:** IM use (20 mg/kg bodyweight, twice): 30 days. SC use (40 mg/kg bodyweight, once): 44 days.

Pig: 18 days.

Milk: Not authorised for use in animals producing milk for human consumption including pregnant animals intended to produce milk for human consumption.

SPECIAL PRECAUTIONS FOR STORAGE

Do not store above 25 °C and protect from moisture. Keep out of the reach and sight of children. Shelf-life after first opening: 28 days.

SPECIAL WARNINGS FOR EACH TARGET SPECIES

None.

SPECIAL PRECAUTIONS FOR USE

This veterinary medicinal product does not contain any antimicrobial preservative.

Special precautions for use in animals: Do not use in piglets of less than 2 kg. Good clinical practices require that the treatment should be based on susceptibility testing of bacteria isolated from diseased animals. If this is not possible, the treatment should be based on local epidemiological information (on the area, farm) of susceptibility of the different strains of bacterial species usually related to the infectious disease. The product should be used taking into account national or local official antimicrobial policies. Special precautions to be taken by the person administering the veterinary medicinal product to animals: The product can cause hypersensitivity (allergy). People with known hypersensitivity to florfenicol, propylene glycol or polyethylene glycols should avoid contact with the veterinary medicinal product. This product contains N-Methylpyrrolidone which may be harmful for the unborn child; therefore, women of child-bearing age must be very careful to avoid exposure via spillage onto the skin or accidental self-injection when administering the product. If you are pregnant, think you may be pregnant or are attempting to conceive, you should not administer the product. Administer the veterinary medicinal product with caution to avoid accidental self-injection. In case of accidental self-injection, seek medical advice immediately and show the package leaflet or the label to the physician. This product may cause skin and eye irritation. Avoid contact with skin or eyes. In case of accidental contact, wash immediately exposed area with plenty of clean water. If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the physician the package leaflet or the label.

Other precautions: Florfenicol is known to be toxic for terrestrial plants and cyanobacteria.

SPECIAL PRECAUTIONS FOR THE DISPOSAL OF UNUSED PRODUCT OR WASTE MATERIALS

Dispose of waste material in accordance with local requirements.

NATURE AND COMPOSITION OF THE IMMEDIATE PACKAGING

Type-II 100 ml glass vial, closed with type-I bromobutyl stopper and aluminium capsule.

FOR ANIMAL TREATMENT ONLY.

TO BE SUPPLIED ONLY ON VETERINARY PRESCRIPTION.

Reg. No. xxxx

Date of last revision: XX/XXXX

Product Registration Holder:

Pahang Pharmacy Sdn Bhd.
Lot 5979 Jalan Teratai, 51/2 Mile off Jalan Meru
41050 Klang, Selangor Darul Ehsan, Malaysia



Manufacturer:
MEVET S.A.U.
Polígono Industrial El segre, p. 409-410
25191 Lleida (Spain)

XXXX-XX/20

